

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Quiz 3  
Semester: Summer 2024

Duration: 25 min  
Full Marks: 15

**CSE 470: Software Engineering**

**Name:**

**ID:**

**Section:**

**Set-A**

At Green Leaf Bookstore, customers can explore and purchase a wide variety of books, magazines, stationery items, and educational supplies. The bookstore offers both physical and digital formats for many of its books. Customers can browse items online through an intuitive web portal or in-store using interactive kiosks. When placing an order, customers select their desired items and specify the quantity. The system automatically calculates the total price of the order, including applicable taxes and any shipping fees.

For items not available immediately due to stock shortages, the order is divided into multiple shipments. The bookstore ensures prompt dispatch of in-stock items and notifies customers about the status of their remaining items, which are shipped as soon as they are restocked. Despite multiple shipments, only a single shipping fee is charged per order. Customers can choose from various payment methods, including credit cards, digital wallets, and store credit.

In addition to purchasing, customers can also subscribe to a yearly membership for exclusive discounts, free shipping, and early access to new releases and special editions. Each time an order or part of an order is ready for shipment, the customer's chosen payment method is charged, with the shipping fee included in the first charge.

To effectively manage their extensive inventory, the bookstore staff keeps meticulous records of both physical and digital stock levels. Every month, a detailed analytics report is generated, highlighting the sales and stock levels of different product categories. This report helps in identifying consumer trends, forecasting demand, and planning inventory procurement. Furthermore, the bookstore occasionally hosts author events and book signings, for which customers can register and pay through the same system. These events are managed through the bookstore's event management module, which tracks attendance, payments, and feedback.

**A. Provide a use-case diagram for the requirements of the Green Leaf Bookstore.**